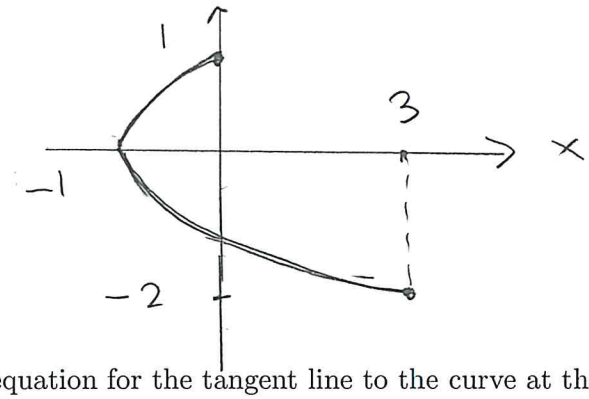
MATH 104 MIDTERM 1 SOLUTION KEY (Spring 2018

Q1) [4+6=10pts] **(Calculus with Parametric Curves)** Given the parametric curve $x = t^2 - 1$, $y = t$ and the parameter interval $-2 \le t \le 1$,

(a) Find the Cartesian equation for the curve, then identify and graph the curve.

Cartesian eq: $x = y^2 - 1$ parabola with $-2 \le y \le 1$



(b) Find an equation for the tangent line to the curve at the point where $t = -1$.

For $t=-1$, $(x_0, y_0) = (0,-1)$

Slope: $\left.\dfrac{dy}{dx}\right|_{(0,-1)} = \left.\dfrac{\frac{dy}{dt}}{\frac{dx}{dt}}\right|_{t=-1} = \left.\dfrac{1}{2t}\right|_{t=-1} = -\dfrac{1}{2} = m$

Eq for the line tangent to the curve

$y - y_0 = m(x - x_0) \Rightarrow y + 1 = -\frac{1}{2}(x - 0) \Rightarrow y = -\frac{x}{2} - 1$

Q2) [8pts] **(Polar Coordinates)** Replace the polar equation $r = -4\sin\theta$ with an equivalent Cartesian equation. Then sketch and identify the curve.

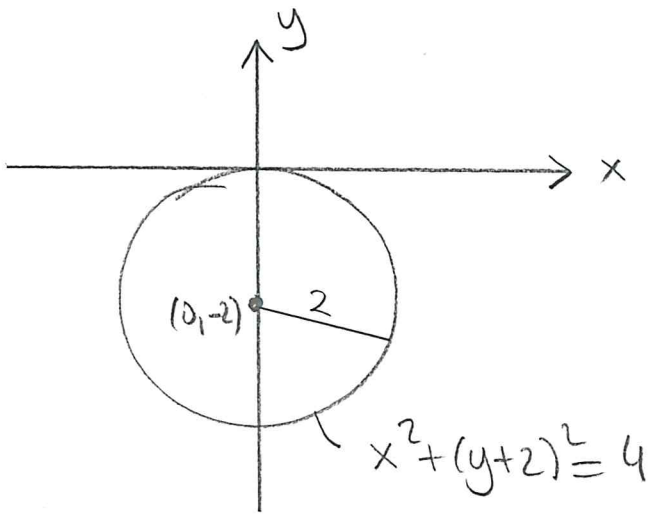
$r^2 = -4r\sin\theta$

$x^2 + y^2 = -4y$

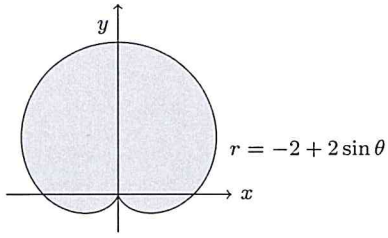
$x^2 + (y+2)^2 = 4$

circle of radius 2

centered at $(0,-2)$



Q3) [12pts] **(Areas and Lengths in Polar Coordinates)** Find the area of the cardioid $r = -2 + 2\sin\theta$ graphed below.



$$A = \frac{1}{2}\int_0^{2\pi} \left[-2(1-\sin\theta)\right]^2 d\theta$$

$$A = \frac{1}{2}\, 4 \int_0^{2\pi} (1 - 2\sin\theta + \sin^2\theta)\, d\theta \qquad \sin^2\theta = \frac{1-\cos(2\theta)}{2}$$

$$A = 2\int_0^{2\pi} \left[1 - 2\sin\theta + \frac{1}{2} - \frac{1}{2}\cos(2\theta)\right] d\theta$$

$$A = 2\left[\frac{3}{2}\theta + \cancel{2\cos\theta}^{\,0} - \cancel{\frac{1}{4}\sin(2\theta)}^{\,0}\right]_0^{2\pi} = 3.2\pi = 6\pi$$

Q4) [3+3+3+3=12pts] **(Vectors and Geometry of Space)** Given the vectors $\boldsymbol{u} = 2\boldsymbol{i} - \boldsymbol{j} - 2\boldsymbol{k}$ and $\boldsymbol{v} = \boldsymbol{j} + 3\boldsymbol{k}$,

(a) Find $\boldsymbol{v} \cdot \boldsymbol{u}$, $|\boldsymbol{u}|$ and $|\boldsymbol{v}|$.

$$\vec{u}\cdot\vec{v} = -1 - 6 = -7 \qquad |\vec{u}| = \sqrt{4+1+4} = 3 \qquad |\vec{v}| = \sqrt{1+9} = \sqrt{10}$$

(b) Find the cosine of the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$.

$$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{|\vec{u}|\,|\vec{v}|} = \frac{-7}{3\cdot\sqrt{10}}$$

(c) Find the vector projection of $\boldsymbol{u}$ onto $\boldsymbol{v}$.

$$\text{proj}_{\vec{v}}\vec{u} = \frac{\vec{u}\cdot\vec{v}}{|\vec{v}|^2}\vec{v} = \frac{-7}{10}(\vec{j} + 3\vec{k})$$

(d) Write a vector $\boldsymbol{w}$ of length $|\boldsymbol{w}| = 3$ in the direction of $\boldsymbol{v}$.

$$\vec{w} = |\vec{w}|\,\frac{\vec{v}}{|\vec{v}|} = 3\cdot\frac{1}{\sqrt{10}}(\vec{j} + 3\vec{k})$$

Q5) [6+4=10pts] **(Vectors and Geometry of Space)** Given the points $P(0,1,2)$, $Q(-2,1,1)$ and $R(1,-3,0)$,

(a) Find $\vec{PQ} \times \vec{PR}$.

$\vec{PQ} = \langle -2, 0, -1 \rangle \quad \vec{PR} = \langle 1, -4, -2 \rangle$

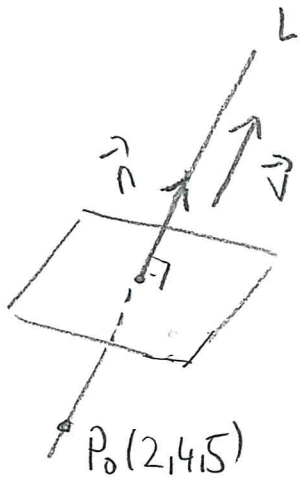
$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 & 0 & -1 \\ 1 & -4 & -2 \end{vmatrix} = \vec{i}(0-4) - \vec{j}(4+1) + \vec{k}(8-0)$$

$$\vec{PQ} \times \vec{PR} = -4\vec{i} - 5\vec{j} + 8\vec{k}$$

(b) Find the area of the triangle determined by the points $P(0,1,2)$, $Q(-2,1,1)$ and $R(1,-3,0)$.

$$A = \frac{1}{2} |\vec{PQ} \times \vec{PR}| = \frac{1}{2}\sqrt{(-4)^2 + (-5)^2 + 8^2} = \frac{1}{2}\sqrt{105}$$

Q6) [8pts] **(Lines and Planes in Space)** Find an equation for the **plane** through the point $P_0(2,4,5)$ perpendicular to the line $x = 5 + t,\ y = 1 + 3t,\ \ z = 4t$.



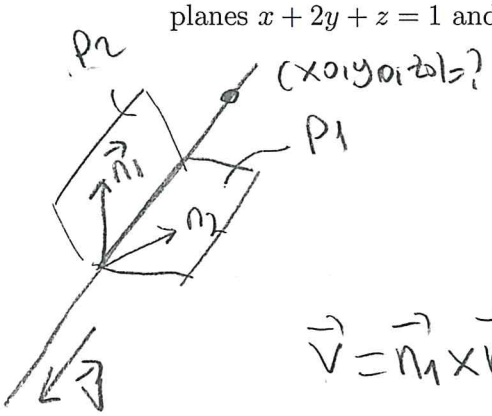
$\vec{v} = \vec{n} = \langle 1, 3, 4 \rangle$

Eqs for the plane through $P_0(2,4,5)$ normal to $\vec{n} = \langle 1,3,4 \rangle$:

$$1(x-2) + 3(y-4) + 4(z-5) = 0$$

$$x + 3y + 4z = 34$$

Q7) [10pts] **(Lines and Planes in Space)** Find parametric equations for the **line** in which the planes $x + 2y + z = 1$ and $x - y + 2z = -8$ intersect.



P1 with $\vec{n}_1 = \langle 1, 2, 1 \rangle$

P2 with $\vec{n}_2 = \langle 1, -1, 2 \rangle$

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ 1 & -1 & 2 \end{vmatrix}$$

$$\vec{v} = \vec{i}(4+1) - \vec{j}(2-1) + \vec{k}(-1-2) = 5\vec{i} - \vec{j} - 3\vec{k}$$

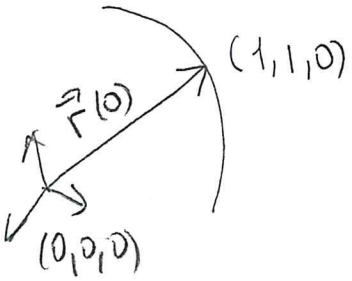
Let (x_0, y_0, z_0) be a point on the line:

$$\left.\begin{aligned} x_0 + 2y_0 + z_0 &= 1 \\ x_0 - y_0 + z_0 &= -8 \end{aligned}\right\} \text{ Let } z_0 = 0. \text{ Then } \left.\begin{aligned} x_0 + 2y_0 &= 1 \\ x_0 - y_0 &= -8 \end{aligned}\right\} \begin{aligned} y_0 &= 3 \\ x_0 &= -5 \end{aligned}$$

Parametric eqs:

$$\begin{aligned} x &= x_0 + tv_1 & y &= y_0 + tv_2 & z &= z_0 + tv_3 \\ x &= -5 + 5t & y &= 3 - t & z &= 0 - 3t \end{aligned}$$

Q8) [8pts] **(Vector-Valued Functions and Motion in Space)** At time $t = 0$, a particle is located at the point $(1, 1, 0)$. The velocity vector of the particle is given by $\boldsymbol{v}(t) = (t^3 + 4t)\boldsymbol{i} + t\boldsymbol{j} + 2t^2\boldsymbol{k}$. Find its position vector $\boldsymbol{r}(t)$.



$$\vec{r}(0) = \langle 1-0, 1-0, 0-0 \rangle = \langle 1, 1, 0 \rangle$$

$$\int \vec{v}(t) = \int (t^3 + 4t)\vec{i} + t\vec{j} + 2t^2\vec{k}$$

$$\vec{r}(t) = \left(\frac{t^4}{4} + 2t^2\right)\vec{i} + \frac{t^2}{2}\vec{j} + \frac{2}{3}t^3\vec{k} + \vec{C}$$

Initial Condition: $\vec{r}(0) = \vec{C} \Rightarrow \vec{C} = \vec{i} + \vec{j}$

Position vector: $\vec{r}(t) = \left(\frac{t^4}{4} + 2t^2 + 1\right)\vec{i} + \left(\frac{t^2}{2} + 1\right)\vec{j} + \frac{2}{3}t^3\vec{k}$

Q9) [4+2+4=10pts] **(Vector-Valued Functions and Motion in Space)** Let

$$\boldsymbol{r}(t) = \sin t\boldsymbol{i} + (t^2 - \cos t)\boldsymbol{j} + e^t\boldsymbol{k}$$

be a position vector of a particle moving in space at time t.

(a) Find the particle's velocity vector $\boldsymbol{v}(t)$ and acceleration vector $\boldsymbol{a}(t)$ at time $t = 0$.

$\boldsymbol{v}(t) = \cos t\,\vec{i} + (2t + \sin t)\vec{j} + e^t\vec{k}$ $\qquad \boldsymbol{v}(0) = \vec{i} + \vec{k}$

$\boldsymbol{a}(t) = -\sin t\,\vec{i} + (2 + \cos t)\vec{j} + e^t\vec{k}$ $\qquad \boldsymbol{a}(0) = 3\vec{j} + \vec{k}$

(b) Find the particle's speed at time $t = 0$.

$|\vec{v}(0)| = \sqrt{1+1} = \sqrt{2}$

(c) Find its unit tangent vector $\mathbf{T}$ at $t = 0$.

$\vec{T}(0) = \dfrac{\vec{v}(0)}{|\vec{v}(0)|} = \dfrac{1}{\sqrt{2}}(\vec{i} + \vec{k})$

Q10) [4+4+4=12pts] **(Arc Length in Space)** Given the curve by the vector function

$$\boldsymbol{r}(t) = 3t\,\boldsymbol{i} + 4\cos t\,\boldsymbol{j} + 4\sin t\,\boldsymbol{k},$$

(a) Find its length over $0 \le t \le \frac{\pi}{2}$.

$\vec{v}(t) = 3\vec{i} - 4\sin t\,\vec{j} + 4\cos t\,\vec{k}$

$|\vec{v}(t)| = \sqrt{9 + 16(\cos^2 t + \sin^2 t)} = 5$

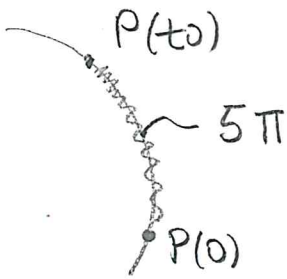
$L = \int_0^{\pi/2} |\vec{v}(t)|\,dt = 5 \cdot \int_0^{\pi/2} dt = 5\frac{\pi}{2}$

(b) Find the arc length parameter $s(t)$ with base point at $t = 0$.

$S(t) = \int_0^t |\vec{v}(\tau)|\,d\tau = 5\int_0^t d\tau = 5t$

(c) Find a point (x_0, y_0, z_0) on the curve at a distance of 5π units along the curve from the point $(0, 4, 0)$.

For $t = 0$, we get $(0, 4, 0)$.



$S(t) = \int_0^t |\vec{v}(\tau)|\,d\tau = 5t$

$S(t_0) = 5\pi \Rightarrow 5t_0 = 5\pi \Rightarrow t_0 = \pi$

For $t_0 = \pi$, $(x_0, y_0, z_0) = (3\pi, -4, 0)$